

Chapter 4: Exploring Algebraic Identities

Complete NCERT Solution & Answer Key | Ganita Manjari Part I (Grade 9 Mathematics)

Exercise Set 4.1 Solutions

Q1(i): Expand $(7x + 4y)^2$

Solution: Using $(a+b)^2 = a^2 + 2ab + b^2$:

$$(7x)^2 + 2(7x)(4y) + (4y)^2 = 49x^2 + 56xy + 16y^2$$

Q1(ii): Expand $(\frac{7}{5}x + \frac{3}{2}y)^2$

$$\text{Solution: } (\frac{7}{5}x)^2 + 2(\frac{7}{5}x)(\frac{3}{2}y) + (\frac{3}{2}y)^2 = \frac{49}{25}x^2 + \frac{21}{5}xy + \frac{9}{4}y^2$$

Q1(iii): Expand $(2.5p + 1.5q)^2$

$$\text{Solution: } (2.5p)^2 + 2(2.5p)(1.5q) + (1.5q)^2 = 6.25p^2 + 7.5pq + 2.25q^2$$

Q1(iv): Expand $(\frac{3}{4}s + 8t)^2$

$$\text{Solution: } (\frac{3}{4}s)^2 + 2(\frac{3}{4}s)(8t) + (8t)^2 = \frac{9}{16}s^2 + 12st + 64t^2$$

Q1(v): Expand $(\frac{x}{y} + \frac{1}{2})^2$

$$\text{Solution: } (\frac{x}{y})^2 + 2(\frac{x}{y})(\frac{1}{2}) + (\frac{1}{2})^2 = \frac{x^2}{y^2} + \frac{x}{y} + \frac{1}{4}$$

Q1(vi): Expand $(\frac{1}{x} + \frac{1}{y})^2$

$$\text{Solution: } (\frac{1}{x})^2 + 2(\frac{1}{x})(\frac{1}{y}) + (\frac{1}{y})^2 = \frac{1}{x^2} + \frac{2}{xy} + \frac{1}{y^2}$$

Q2(i): Find $(64)^2$ using identity

$$\text{Solution: } (60 + 4)^2 = 60^2 + 2(60)(4) + 4^2 = 3600 + 480 + 16 = 4096$$

Q2(ii): Find $(105)^2$ using identity

$$\text{Solution: } (100 + 5)^2 = 100^2 + 2(100)(5) + 5^2 = 10000 + 1000 + 25 = 11025$$

Q2(iii): Find $(205)^2$ using identity

$$\text{Solution: } (200 + 5)^2 = 200^2 + 2(200)(5) + 5^2 = 40000 + 2000 + 25 = 42025$$

Exercise Set 4.2 Solutions

Q1(i): Factor completely: $9x^2 + 24xy + 16y^2$

$$\text{Solution: } (3x)^2 + 2(3x)(4y) + (4y)^2 = (3x + 4y)^2$$

Q1(ii): Factor completely: $4s^2 + 20st + 25t^2$

$$\text{Solution: } (2s)^2 + 2(2s)(5t) + (5t)^2 = (2s + 5t)^2$$

Q1(iii): Factor completely: $49x^2 + 28xy + 4y^2$

$$\text{Solution: } (7x)^2 + 2(7x)(2y) + (2y)^2 = (7x + 2y)^2$$

Q1(iv): Factor: $\frac{64}{9}p^2 + \frac{32}{3}pq + 4q^2$

$$\text{Solution: } (\frac{8}{3}p)^2 + 2(\frac{8}{3}p)(2q) + (2q)^2 = (\frac{8}{3}p + 2q)^2$$

Q1(v): Factor: $3a^2 + 4ab + \frac{4}{3}b^2$

$$\text{Solution: Take } \frac{1}{3} \text{ common: } \frac{1}{3}(9a^2 + 12ab + 4b^2) = \frac{1}{3}(3a + 2b)^2$$

Q1(vi): Factor: $\frac{9}{5}s^2 + 6sv + 5v^2$

Solution: Take $1/5$ common: $1/5 (9s^2 + 30sv + 25v^2) = 1/5 (3s + 5v)^2$

Q2(i): Evaluate $(79)^2$ using $(a-b)^2$

Solution: $(80 - 1)^2 = 80^2 - 2(80)(1) + 1^2 = 6400 - 160 + 1 = 6241$

Q2(ii): Evaluate $(193)^2$ using $(a-b)^2$

Solution: $(200 - 7)^2 = 200^2 - 2(200)(7) + 7^2 = 40000 - 2800 + 49 = 37249$

Q2(iii): Evaluate $(299)^2$ using $(a-b)^2$

Solution: $(300 - 1)^2 = 300^2 - 2(300)(1) + 1^2 = 90000 - 600 + 1 = 89401$

Exercise Set 4.3 Solutions

Q1(i): Evaluate $(117)^2$

Solution: $(100 + 10 + 7)^2 = 100^2 + 10^2 + 7^2 + 2(1000 + 700 + 70) = 10000 + 100 + 49 + 3540 = 13689$

Q1(ii): Evaluate $(78)^2$

Solution: $(80 - 2)^2 = 6400 - 320 + 4 = 6084$

Q1(iii): Evaluate $(198)^2$

Solution: $(200 - 2)^2 = 40000 - 800 + 4 = 39204$

Q1(iv): Evaluate $(214)^2$

Solution: $(200 + 14)^2 = 40000 + 5600 + 196 = 45796$

Q1(v): Evaluate $(1104)^2$

Solution: $(1100 + 4)^2 = 1210000 + 8800 + 16 = 1218816$

Q1(vi): Evaluate $(1120)^2$

Solution: $(1100 + 20)^2 = 1210000 + 44000 + 400 = 1254400$

Q2(i): Factor $16y^2 - 24y + 9$

Solution: $(4y)^2 - 2(4y)(3) + 3^2 = (4y - 3)^2$

Q2(ii): Factor $9/4 s^2 + 6st + 4t^2$

Solution: $(3/2 s)^2 + 2(3/2 s)(2t) + (2t)^2 = (3/2 s + 2t)^2$

Q2(iii): Factor $m^2 + 2/9 mk + 1/3 nk + 2/3 mn + n^2 + k^2/16$

Solution: $(m + n + k/4)^2 = (m + n + k/4)^2$

Q2(iv): Factor $p^2 - p/2 + 1/16$

Solution: $(p - 1/4)^2 = (p - 1/4)^2$

Q2(v): Factor $9a^2 + 4b^2 + c^2 - 12ab + 6ac - 4bc$

Solution: $(-3a + 2b + c)^2$ or $(3a - 2b - c)^2$

Q3(i): Expand $(p + 3q + 7r)^2$

Solution: $p^2 + 9q^2 + 49r^2 + 2(p)(3q) + 2(3q)(7r) + 2(7r)(p) = p^2 + 9q^2 + 49r^2 + 6pq + 42qr + 14rp$

Q3(ii): Expand $(3x - 2y + 4z)^2$

Solution: $(3x)^2 + (-2y)^2 + (4z)^2 + 2(3x)(-2y) + 2(-2y)(4z) + 2(4z)(3x) = 9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24zx$

Q4: Verify identity: $(a+b-c)^2 + (a-b+c)^2 + (-a+b+c)^2 + (a+b+c)^2$

Solution: Summing expanded terms, all cross terms canceling in pairs yields $4(a^2 + b^2 + c^2)$. Yes, it is an identity.

Exercise Set 4.4 Solutions

Q1(i): Fill: $s^2 - 11s + 24 = (\quad)(\quad)$

Solution: $a+b = -11$, $ab = 24 \Rightarrow a = -3$, $b = -8$. Result: $(s - 3)(s - 8)$

Q1(ii): Fill: $(\quad)(x + 1) = 3x^2 - 4x - 7$

Solution: $3x^2 - 4x - 7 = (3x - 7)(x + 1)$. Result: $(3x - 7)$

Q1(iii): Fill: $10x^2 - 11x - 6 = (2x - \quad)(\quad + 2)$

Solution: $10x^2 - 11x - 6 = (2x - 3)(5x + 2)$. Result: $(2x - 3)(5x + 2)$

Q1(iv): Fill: $6x^2 + 7x + 2 = (\quad)(\quad)$

Solution: $6x^2 + 4x + 3x + 2 = (2x + 1)(3x + 2)$

Q2(iii): Evaluate 23×17 **using identity**

Solution: $(20 + 3)(20 - 3) = 20^2 - 3^2 = 400 - 9 = 391$

Q2(vi): Evaluate 18×29 **using identity**

Solution: $(20 - 2)(30 - 1) = 600 - 20 - 60 + 2 = 522$

Q3(i): Factor $9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$

Solution: $(-3a + b + 2c)^2$ or $(3a - b - 2c)^2$

Q3(ii): Factor $16s^2 + 25t^2 - 40st$

Solution: $(4s)^2 - 2(4s)(5t) + (5t)^2 = (4s - 5t)^2$

Q3(iii): Factor $r^2 - r - 42$

Solution: Split -1 into -7 and +6: $(r - 7)(r + 6)$

Q3(iv): Factor $49g^2 + 14gh + h^2$

Solution: $(7g)^2 + 2(7g)(h) + h^2 = (7g + h)^2$

Q3(v): Factor $64u^2 + 121v^2 + 4w^2 - 176uv - 32uw + 44vw$

Solution: $(-8u + 11v + 2w)^2$ or $(8u - 11v - 2w)^2$

Exercise Set 4.5 Solutions

Q1(i): Simplify: $(3p^2 - 3pq - 18q^2) / (p^2 - pq - 10q^2)$

Solution: $3(p - 3q)(p + 2q) / [(p - 5q)(p + 2q)] = 3(p - 3q) / (p - 5q)$

Q1(ii): Simplify: $(n^3 - n^2m - nm^2 + m^3) / (5m^2 - 10mn + 5n^2)$

Solution: $(n - m)^2(n + m) / [5(n - m)^2] = (n + m) / 5$

Q1(iv): Simplify: $(4y^2 - 20yz + 25z^2) / (25z^2 - 4y^2)$

Solution: $(2y - 5z)^2 / [(5z - 2y)(5z + 2y)] = -(2y - 5z) / (2y + 5z)$ or $(5z - 2y) / (2y + 5z)$

Q1(vi): Simplify: $(p^4 - 16) / (p^2 - 4p + 4)$

Solution: $(p^2 + 4)(p + 2)(p - 2) / (p - 2)^2 = (p^2 + 4)(p + 2) / (p - 2)$

End-of-Chapter Exercises Solutions

Q1(i): $(-3x + 4)^2$

Solution: $(-3x)^2 + 2(-3x)(4) + 4^2 = 9x^2 - 24x + 16$

Q1(ii): $(2s + 7)(2s - 7)$

Solution: $(2s)^2 - 7^2 = 4s^2 - 49$

Q1(iii): $(p^2 + 1/2)(p^2 - 1/2)$

Solution: $(p^2)^2 - (1/2)^2 = p^4 - 1/4$

Q1(v): $(s - 2t)(s^2 + 2st + 4t^2)$

Solution: Using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$: $s^3 - 8t^3$

Q1(viii): $(x - 1/3 y)^3$

Solution: $x^3 - 3x^2(1/3 y) + 3x(1/9 y^2) - 1/27 y^3 = x^3 - x^2y + 1/3 xy^2 - 1/27 y^3$

Q2(ii): 104 x 96 using identity

Solution: $(100 + 4)(100 - 4) = 100^2 - 4^2 = 10000 - 16 = 9984$

Q2(v): 199^3 using identity

Solution: $(200 - 1)^3 = 200^3 - 3(200)^2(1) + 3(200)(1)^2 - 1 = 8000000 - 120000 + 600 - 1 = 7880599$

Q3(i): Factor $4y^2 + 1/16 y^2 + 1$

Solution: $(2y + 1/(4y))^2 - 2(2y)(1/(4y)) + 1 = (2y + 1/(4y))^2$ or $(2y + 1/(4y))^2$

Q3(iii): Factor $27b^3 - 1/(64b^3)$

Solution: $(3b - 1/(4b))(9b^2 + 3/4 + 1/(16b^2))$

Q3(vii): Factor $p^3 + q^3 + r^3/27 - 3pqr$

Solution: Using identity: $(p + q + r/3)(p^2 + q^2 + r^2/9 - pq - qr/3 - rp/3)$

Q7: Area of path around 40m square playground with path width s

Solution: Outer side = $40 + 2s$. Outer Area = $(40 + 2s)^2$. Inner Area = $40^2 = 1600$.

Path Area = $(1600 + 160s + 4s^2) - 1600 = 4s^2 + 160s \text{ m}^2$

Q8: If $x + 1/x = 10/3$, find x

Solution: $3x^2 - 10x + 3 = 0 \Rightarrow (3x - 1)(x - 3) = 0 \Rightarrow x = 3 \text{ or } x = 1/3$

Q9: Pool area = $2x^2 + 7x + 3$, width = $2x + 1$. Find length.

Solution: $2x^2 + 7x + 3 = (2x + 1)(x + 3)$. Length = $x + 3$ hastas

Q10: If $x-2$ and $x-1/2$ are factors of $px^2 + 5x + r$, show $p = r$

Solution: $p(2)^2 + 5(2) + r = 0 \Rightarrow 4p + r = -10$.

$p(1/2)^2 + 5(1/2) + r = 0 \Rightarrow p/4 + r = -5/2 \Rightarrow p + 4r = -10$.

Comparing gives $4p + r = p + 4r \Rightarrow 3p = 3r \Rightarrow p = r$. (Proved)

Q11: If $a+b+c=5$, $ab+bc+ca=10$, prove $a^3+b^3+c^3-3abc = -25$

Solution: $a^2+b^2+c^2 = (a+b+c)^2 - 2(ab+bc+ca) = 25 - 20 = 5$.

Identity: $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2 - (ab+bc+ca)) = 5(5 - 10) = 5(-5) = -25$. (Proved)

Q12: Check $n^3 - n$ is divisible by 6 for natural numbers n

Solution: $n^3 - n = n(n-1)(n+1) = (n-1)n(n+1)$, the product of 3 consecutive integers. At least one is even and exactly one is divisible by 3, so the product is **always divisible by 6**.

Q13(i): Find $x^3 + y^3 - 12xy + 64$ when $x + y = -4$

Solution: $x + y + 4 = 0 \Rightarrow x^3 + y^3 + 4^3 - 3(x)(y)(4) = 0 \Rightarrow x^3 + y^3 + 64 - 12xy = 0$